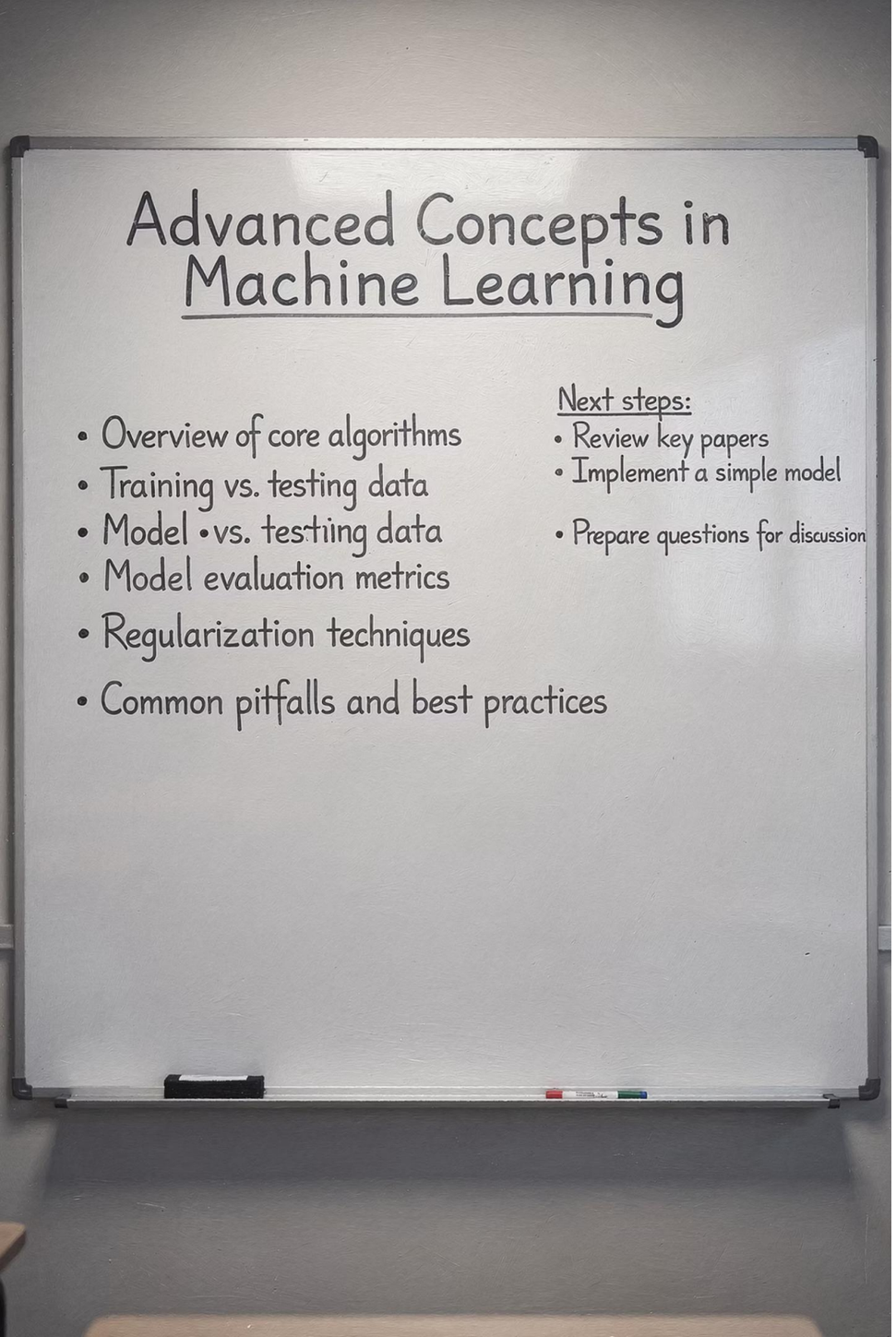
Business Analytics with R – Regression Analysis



Outline

01

Goals of this lecture

02

What is a (linear) regression analysis?

03

How to perform a regression analysis in R?

Goals of the Current Lecture

OLS Principles

Students know the basic principles of OLS regression analysis

Estimation & Interpretation

Estimating and interpreting Linear Regression (Ordinary Least Squares)

Predictions

Can use the estimated regression model to make predictions

Model Quality

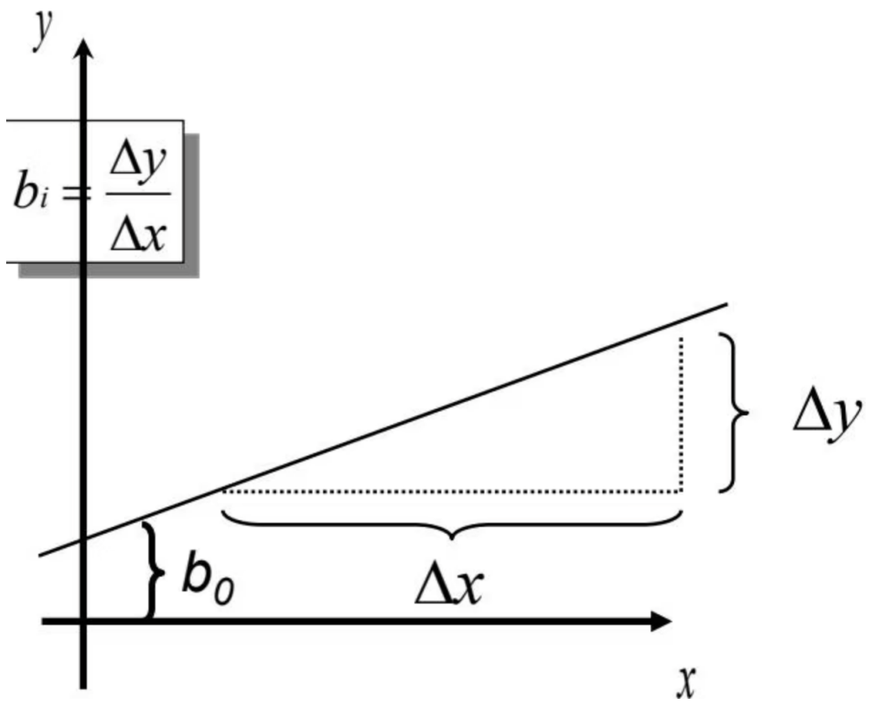
Can evaluate the quality of model predictions

Regression Diagnostic

Know how to perform a regression diagnostic

Students can apply regression analyses in R across all of the above areas.

What is Linear Regression?



Definition

A way of predicting the value of one variable from another.

It is a hypothetical model of the relationship between two variables.

The model used is a linear one.

Therefore, we describe the relationship using the equation of a straight line.

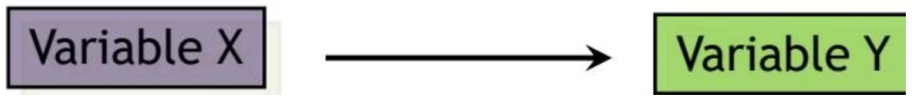
The General Equation

$$Y_i = b_0 + b_i X_i + \varepsilon_i$$

- ***b_i*** — Regression coefficient for the predictor; Gradient (slope) of the regression line; Direction/Strength of Relationship
- ***b_0*** — Intercept (value of Y when *X = 0*); Point at which the regression line crosses the Y-axis (ordinate)

Linear Regression (OLS)

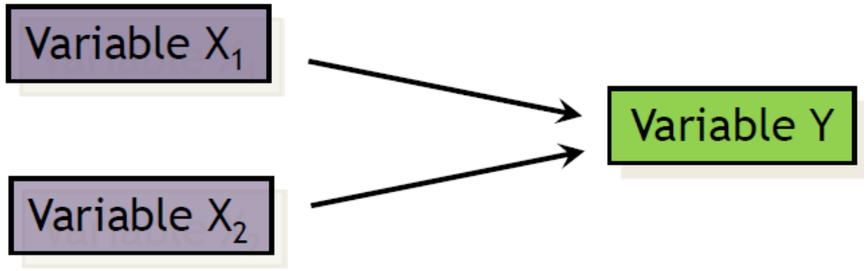
SIMPLE LINEAR REGRESSION



One independent Variable:

$$Y = f(X)$$

MULTIPLE LINEAR REGRESSION



Multiple independent variables:

$$Y = f(X_1, X_2, \dots, X_i, \dots, X_J)$$

Regression Model: We Make Errors

Full Model (with error)

$$Y_i = b_0 + b_i X_i + \varepsilon_i$$

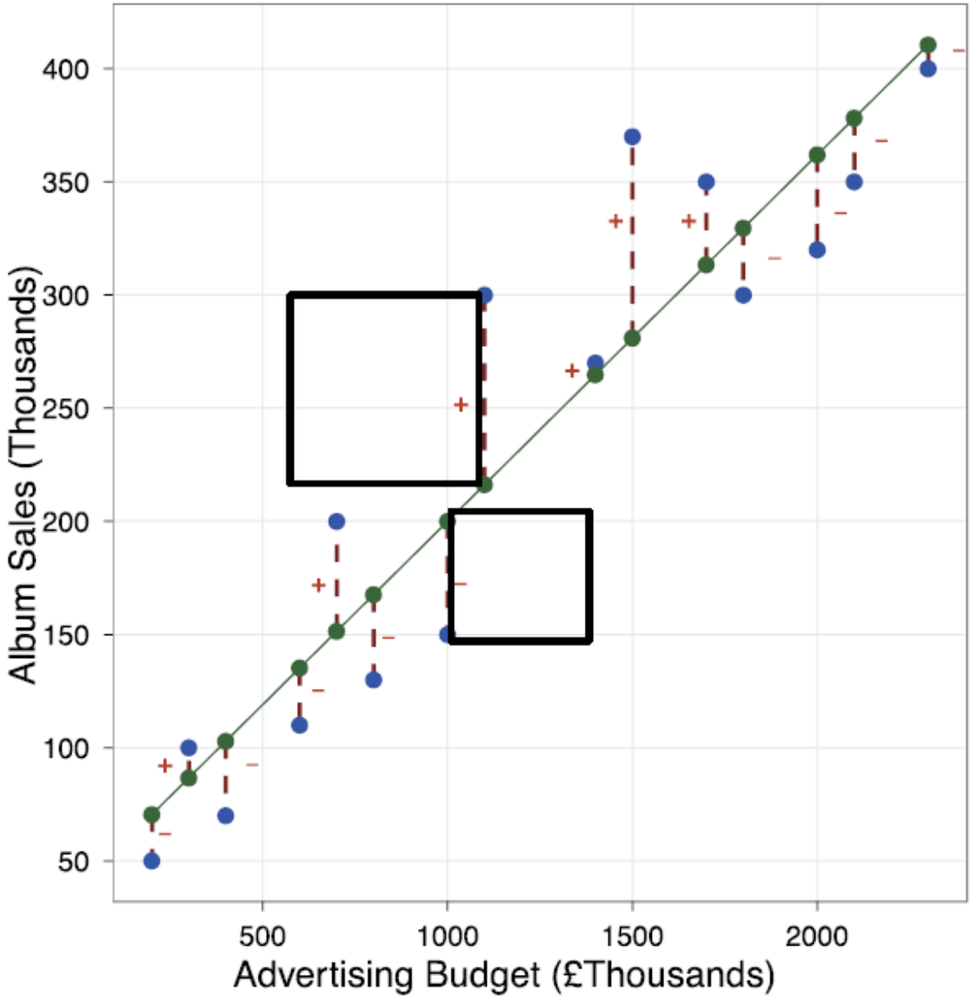
Error / Residual: ε_i

Predicted Value

$$\hat{Y}_i = b_0 + b_i X_i$$

Predicted value of dependent variable

The difference between the observed value Y_i and the predicted value $\hat{Y}_i$ is the residual (error) ε_i.



The Method of Least Squares

FIGURE 8.4 — A scatterplot of some data with a line representing the general trend. The vertical lines (dotted) represent the differences (or residuals) between the line and the actual data.

The axes show **Advertising Budget (£Thousands)** on the x-axis and **Album Sales (Thousands)** on the y-axis. Data points are shown as blue circles, the green line represents the linear trend, and vertical dotted red lines connect each data point to the regression line, representing the residuals.

How Well Does the Model Fit the Data?

There are two ways to assess the accuracy of the model in the sample:

Residual Statistics & Influence/Leverage Diagnostics

- Standardized Residuals
- Influential cases
- Cook's distance

Model Fit Metrics

- **R** — The correlation between the observed values of the outcome, and the values predicted by the model
- **R^2** — The proportion of variance accounted for by the model
- **Adj. R^2** — Controlled for the number of predictors

R and *R^2*

R

The correlation between the observed values of the outcome, and the values predicted by the model.

R^2

The proportion of variance accounted for by the model.

Adj. R^2

Controlled for the number of predictors – R^2 *increases with an increasing number of predictors*

Estimating a Regression Model in R

Now we estimate a multivariate model with two predictors.

```
model <- lm(totalRent ~ livingSpace +
yearConstructed + balcony, data = df)
```

Displaying the model with `stargazer`:

```
stargazer(model, type = "text")
```

Regression Output: stargazer Results

	Dependent variable:
	totalRent
livingSpace	5.729***
	(0.082)
yearConstructed	0.015
	(0.028)
balconyy	13.160***
	(2.904)
Constant	51.512
	(55.697)
Observations	1,096
R2	0.822
Adjusted R2	0.822
Residual Std. Error	43.701 (df = 1092)
F Statistic	1,681.918*** (df = 3; 1092)
Note:	*p<0.1; **p<0.05; ***p<0.01

Predictions on Test Data

Predicting for test data and storing predictions as new variable in test data frame:

```
test$prediction <- predict(model3, newdata= test)
```

```
test$prediction
```

The `predict()` function applies the estimated model to new (test) data, generating predicted values of `totalRent` for each observation in the test set.

Measures of Model Accuracy

```
##Calculating measures of model accuracy

###calculating mape
test <- test %>%
  mutate(MAPE = abs(prediction - totalRent)/totalRent)

mean(test$MAPE)

###Calculating Sales Ratios
test <- test %>%
  mutate(SR = prediction / totalRent)

mean(test$SR)
```

MAPE

Mean Absolute Percentage Error — measures average prediction error as a proportion of actual values

Sales Ratio (SR)

Ratio of predicted to actual value — indicates systematic over- or under-prediction

Model Validation & Performance Generalisation

To ensure a model generalises well beyond the training data, several validation strategies can be applied.

1

Holdout Validation

Split data once into Training and Test sets

2

Random Sub-sampling

Repeat random splits across multiple partitions

3

k-Fold Cross Validation

Rotate test fold across k equal partitions

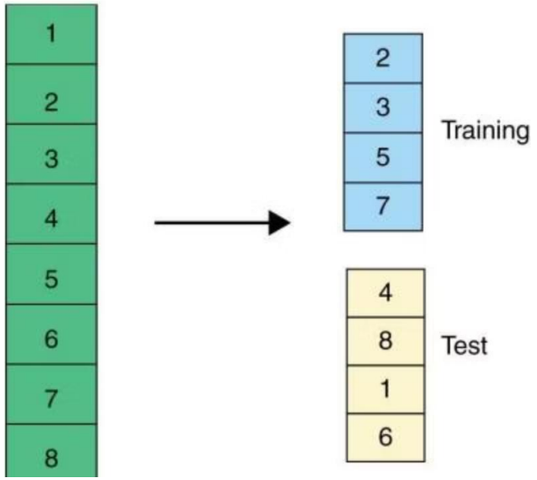
4

Leave-One-Out

Each observation serves as the test set once

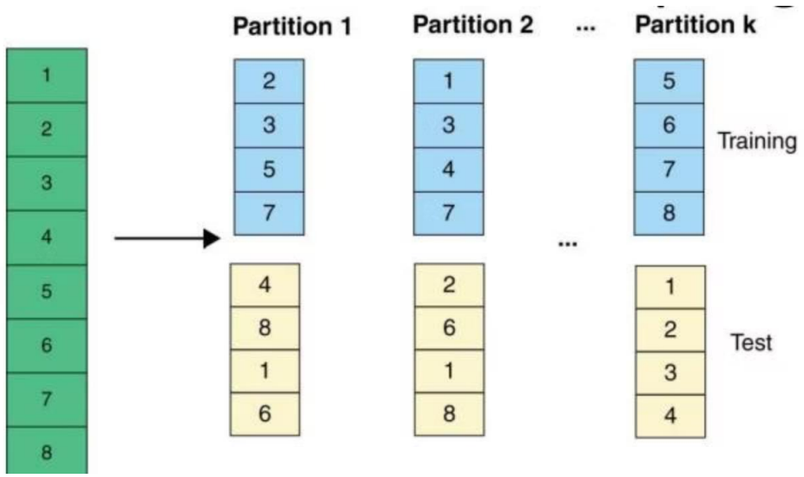
Holdout Validation & Random Sub-sampling

Holdout Validation



The full dataset is split into a **Training** set (light blue) and a **Test** set (light yellow). The model is trained once and evaluated once.

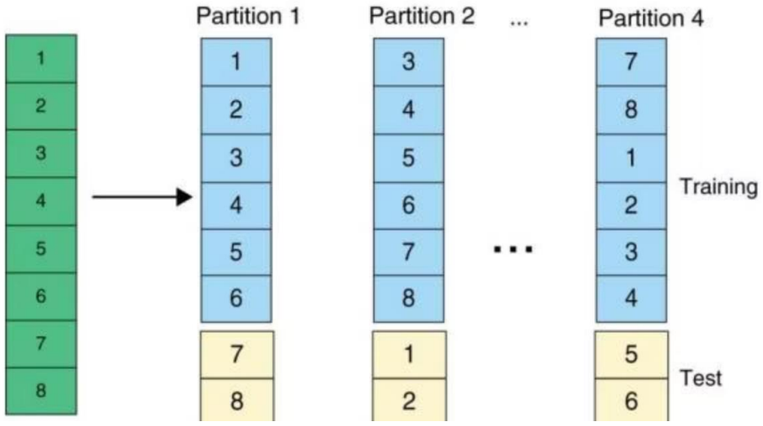
Random Sub-sampling



The data is randomly split into Training and Test sets multiple times across **k partitions**, and results are averaged across all partitions.

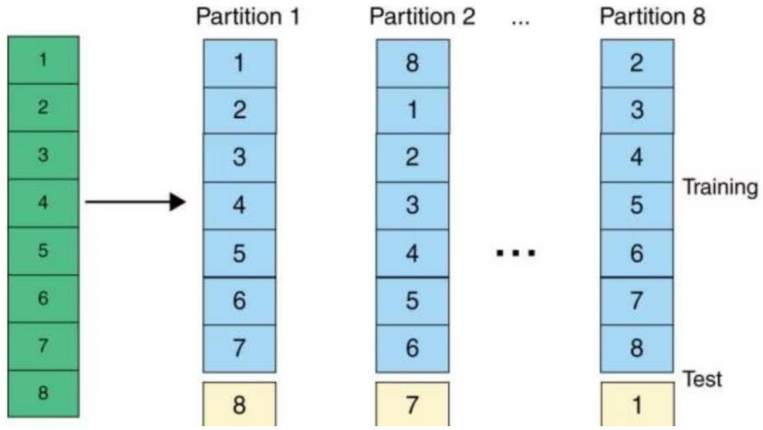
k-Fold Cross Validation & Leave-One-Out

k-Fold Cross Validation



Data is divided into k equal folds. In each iteration, one fold serves as the **Test** set (yellow) and the remaining k−1 folds serve as the **Training** set (blue).

Leave-One-Out



Each individual observation serves as the **Test** set exactly once, while all remaining observations form the **Training** set. This is repeated for every observation in the dataset.

k-Fold Cross Validation in R

```
library(caret)

# Define number of folds
k <- 3

# Create folds
folds <- createFolds(df.numeric.variables$totalRent, k = 3, list = FALSE)

for(i in 1:k){
 # We keep fold i for testing the model and use all other folds for training the model.
 test <- df.numeric.variables%>%
 filter(folds==i)
 train <- df.numeric.variables%>%
 filter(folds!=i)

 # Training model
 model <- lm(totalRent~livingSpace+yearConstructed,data=train)
 cat("Model summary for fold",i,":\n")
 print(summary(model))
}
```

Summary: Key Concepts Covered

1 Basic principles of OLS regression analysis

Understanding the linear model equation $Y_i = b_0 + b_1 X_i + \varepsilon_i$, slope, intercept, and residuals

2 Estimating and interpreting regression in R

Using `lm()` and `stargazer()` to fit and display multivariate models

3 Predictions and model accuracy

Using `predict()`, MAPE, and Sales Ratios to evaluate model performance on test data

4 Model validation strategies

Holdout, random sub-sampling, k-fold cross validation, and leave-one-out using `caret`

Questions?

OLS Regression

Basic principles, slope, intercept, and the regression equation

R Implementation

Estimating models with `lm()`, displaying with `stargazer()`

Model Accuracy

MAPE, Sales Ratios, R^2, and Adjusted R^2

Validation

Holdout, k-fold cross validation, and leave-one-out strategies

Business Analytics with R – Regression Analysis Thank you for your attention!